Heayoun (Cindy) Choi

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Research Interests

Generalization and Autonomous Adaptation: Developing systems that leverage prior knowledge and limited new experience to adapt beyond training distributions and continuously acquire missing knowledge in novel environments.
Reasoning in the Physical World: Combining learned representations with structured reasoning and physical constraints for more reliable decision-making in real-world settings.
Applications: Complex systems on Earth, autonomous exploration, and sustainable habitats beyond Earth.

Education

M.S. in Computer Science & Engineering at Chung-Ang University — Sep 2024 – Aug 2026
AI and Robotics Lab (Advisor: Eunwoo Kim) | [Lab] — *Seoul, Korea*

B.S. in Business Administration at Soongsil University — Feb 2018 – Feb 2022
Summa Cum Laude — *Seoul, Korea*

Publications

PUBLISHED

XIL: Cross-Expanding Incremental Learning | [Paper] [Code] — **2026**
Heayoun Choi, Hyundong Jin, and Eunwoo Kim — *ICLR 2026*

Detecting Failures in Robot Manipulation via Distance-Aware Temporal Modelling | [Paper]
Heayoun Choi and Eunwoo Kim — *Electronics Letters*

Relevance-Guided Retrieval under Continually Expanding Knowledge Bases | [Paper] — **2025**
Hyundong Jin, Heayoun Choi, and Eunwoo Kim — *ICTC 2025*

UNDER REVIEW

Sparse Feature Unlearning Mitigates State Hallucination in Vision-Language-Action Models — **2026**
Jiho Lee, Taekyung Kim, Heayoun Choi, Jeongeun Park, and Eunwoo Kim

Can Vision-Language-Action Models Adapt to Visual Perturbations Using a Test-Time Trajectory?
Jiho Lee, Heayoun Choi, and Eunwoo Kim

Projects

FUNDED RESEARCH

Socially-Aligned and Trustworthy Robot Learning | *2026*

- Selected for National IT Industry Promotion Agency (NIPA)'s Advanced GPU Utilization Support Program, receiving 2 NVIDIA B200 GPU servers (16 GPUs); proposed the research direction and led the successful proposal development.

Lifelong Robot Learning: Foundation and Continuous Adaptation to Tasks, Domains, and Modalities | *2026*

- Selected for a three-year National Research Foundation of Korea (NRF) Core Research (Type B) grant of up to USD 145,000 per year; proposed the research direction and led the successful proposal development.

SOFTWARE PROJECT

Gachimind | *2023* | [GitHub]

- Developed a real-time video mock interview platform using React.js, Node.js, WebRTC, and Socket.io, including session management, REST APIs, and authentication.
- Built the frontend with React.js, Redux, RTK Query, and Styled Components, using Git and Figma for collaborative development and UI/UX design.

Technical Skills

Robotics ROS/ROS 2, MuJoCo, Franka Emika Panda & Franka Research 3 (FR3), Real-Robot Manipulation & Experimentation

Programming Python, C/C++, JavaScript/TypeScript, SQL, VBA, nesC

Teaching Experience

Teaching Assistant at Chung-Ang University 2025 – 2026
Machine Learning (CSE54616) | *Capstone Design (CSE56120)* *Seoul, Korea*

Academic Service

Conference and Journal Reviewer 2025 – 2026
NeurIPS | *ECCV* | *ICCV* | *CoRL* | *AAAI* | *Neural Networks*

Honors and Awards

CAU Graduate Research Scholarship (GRS) 2024 – 2026
Chung-Ang University

Best Corporate Award at Hang-Hackathon Oct 2023
EcubeLab – "AngleEye," Computer Vision System for Precision Angle Estimation

Academic Excellence Award Feb 2022
Soongsil University

Best Award for Model United Nations Feb 2021
Korean National Diplomatic Academy

Best Award for Public Diplomacy Video Feb 2021
Korean National Diplomatic Academy

Work Experience

FNPricing Mar 2023 – Sep 2023
Fixed Income Valuation Analyst *Seoul, Korea*

- Developed and refined a bond evaluation engine to improve the accuracy and efficiency of pricing and assessment workflows.
- Managed and analyzed financial datasets and evaluated a range of instruments, including Mortgage-Backed Securities (MBS) and government bonds.

Yonhap Infomax Mar 2022 – Sep 2022
Financial Engineering Intern *Seoul, Korea*

- Maintained and validated bond-related datasets and financial calculation systems.

Certifications

Advanced Data Analytics Semi-Professional (ADsP) | *Certification demonstrating expertise in data analytics*

Computerized Accounting Operator Level 2 | *Certified proficiency in computerized accounting systems and operations*